

SynthProof — What the Numbers Mean, and How We Compare

This report reads every figure from our **full-dataset** runs (every row of four datasets) and says, in plain language, what each number is telling you — then sets it honestly against the eleven papers we build on. Short version: on the numbers, our synthetic data is **as good as the real data for analysis** while leaking **no detectable membership**; and on contribution, we ship the one thing none of them do — a **signed release a stranger can verify from the artifact alone**.

1 · How to read these numbers

Five plain-English explainers — what each metric measures, what a good value looks like, and why it matters.

Correlation error

How faithfully the synthetic data keeps the relationship between two columns (e.g. age and income). We compare the correlation in the synthetic data to the correlation in the real data.

Good value: 0 = identical to real; below ~0.02 is near-perfect; above ~0.1 means the relationship is distorted.

So what: It tells you the synthetic data still has the real patterns an analyst would study.

TSTR F1 vs real-data ceiling

Train a model on the SYNTHETIC data, then test it on REAL data (TSTR). Divide by the score of a model trained on the real data (TRTR = the ceiling).

Good value: Close to 100% = the synthetic data is as useful as the real, private data for building models.

So what: It proves the synthetic data can replace the sensitive data for actual machine learning — the whole point.

Proved ϵ (epsilon)

The privacy budget mathematically charged for the release. Lower ϵ = stronger privacy.

Good value: Whatever you asked for or less — 'calibration never overspends' means we never charge more than requested.

So what: It is the guarantee itself: a smaller ϵ means an attacker learns less about any one person.

Audited ϵ / GDP μ

An empirical attack's estimate of how much actually leaked. The standard canary audit reads 0.000 — but that is the tool's DETECTION FLOOR, not proof of zero.

Good value: A number reported WITH its detection limit. Ours: canary=0 (floor), GDP $\mu \approx 0.15$ –0.36 (a real signal).

So what: Being honest that '0' means 'the instrument can't see leakage this small', not 'nothing leaked', is itself a contribution.

Membership-inference AUC

Can an attacker guess whether one specific person was in the training data? Measured as AUC.

Good value: 0.5 = a coin flip = safe. Ours stays within ~0.05 of 0.5 everywhere.

So what: It is the direct 'can you re-identify me?' test — and the answer is no better than chance.

2 · What our full-dataset results actually say

Every row of each dataset. Best operating point shown; read each bullet as a sentence.

UCI Adult CENSUS

30,162 rows (full)

- **Relationships preserved:** the age vs hours-worked correlation is reproduced to within **0.0072** error (best: aim at $\epsilon=2$) — essentially indistinguishable from real.
- **Useful for ML:** a model trained on our synthetic data reaches **92%** of real-data accuracy (TSTR F1 0.642 vs real-data 0.696; best: mst).
- **Privacy holds:** the membership-inference attacker stays within 0.034 of a coin-flip (0.5) — no detectable leak about who was in the data.

ACSIIncome CA-2018 CENSUS

195,665 rows (full)

- **Relationships preserved:** the age vs hours-worked correlation is reproduced to within **0.0017** error (best: pairwise at $\epsilon=8$) — essentially indistinguishable from real.
- **Useful for ML:** a model trained on our synthetic data reaches **89%** of real-data accuracy (TSTR F1 0.676 vs real-data 0.759; best: mst).
- **Privacy holds:** the membership-inference attacker stays within 0.015 of a coin-flip (0.5) — no detectable leak about who was in the data.

UCI Bank Marketing FINANCE

45,211 rows (full)

- **Relationships preserved:** the age vs account balance correlation is reproduced to within **0.0009** error (best: pairwise at $\epsilon=4$) — essentially indistinguishable from real.
- **Useful for ML:** a model trained on our synthetic data reaches **90%** of real-data accuracy (TSTR F1 0.492 vs real-data 0.545; best: aim).
- **Privacy holds:** the membership-inference attacker stays within 0.037 of a coin-flip (0.5) — no detectable leak about who was in the data.

UCI Diabetes 130 HEALTHCARE

99,492 rows (full)

- **Relationships preserved:** the days in hospital vs #medications correlation is reproduced to within **0.1185** error (best: mst at $\epsilon=0.5$) — partly distorted.
- **Useful for ML:** a model trained on our synthetic data reaches **99%** of real-data accuracy (TSTR F1 0.487 vs real-data 0.490; best: mst).
- **Privacy holds:** the membership-inference attacker stays within 0.052 of a coin-flip (0.5) — no detectable leak about who was in the data.

Reading it together: on all four datasets our synthetic data keeps the real relationships (correlation error as low as **0.0009** on Bank), stays useful for machine learning (up to **92%** of real-data accuracy on Adult), and shows **no membership leak** (attacker $\approx$ coin-flip). The privacy audit reads audited $\epsilon=0$ with the canary tool (its detection floor) and a real GDP signal — independent $\mu=0.363$ · aim $\mu=0.183$ · mst $\mu=0.151$ — reported honestly as a floor, not as 'zero leakage'.

3 · How we compare to the literature — honestly

The exact relationship to each of the 11 papers we build on (research/18, verified). We are careful: on the mechanisms we *match* the state of the art; where we go beyond is the checkable, signed artifact.

#	Paper	What they do	Verdict	Why — in one line
P10	Dibia — DP Privacy Label PoPETs'26	Designed a 9-field privacy label for DP releases — but only on paper, unsigned, with no way to report an audit's limits.	WE GO FURTHER	They propose the label; we BUILD it, add an Ed25519 signature and an operating-range field, and make conformance = passing our checker. Their own expert called the missing pieces 'privacy theater' — we supply them.
P11	Song — Mental Models CSCW'24	Showed practitioners do not verify DP guarantees — they trust blindly.	WE ANSWER IT	They diagnose the problem and build no tool. Our boundary-audit is the tool a third party can actually run — turning blind trust into a check.
P9	Abowd — Census TopDown HDSR'22	A real national deployment that publishes exact 'invariants' outside the privacy budget — documented only in prose.	WE MAKE IT CHECKABLE	Their invariants live in a PDF a human must read; RB8 + RB11-14 turn them into a machine-checkable field a script verifies.
P4	Ganev — Domain Extraction ICLR'25	Proved that reading a column's min/max from private data silently breaks DP.	WE MAKE IT CHECKABLE	They diagnose the leak; RB6 turns 'how was the domain obtained?' into a value a checker reads (declared/codebook/charged vs inferred).
P5	Ganev — Discretization CCS'25	Proved data-derived bins leak the input distribution.	WE MAKE IT CHECKABLE	RB9 flags data-derived binning as a leak from the artifact alone — their finding becomes an automatic check.
P2	Annamalai — Tight Auditing USENIX'24	Showed black-box audits of DP synthetic data read $\epsilon \approx 0$ even when the true ϵ is 4.	WE MATCH + STAY HONEST	We reproduce this (our GDP audit) and, crucially, REPORT it: our audited $\epsilon=0$ is labelled a black-box floor, not 'no leakage'. Most work hides this.
P3	Cebere — Grey-Box Auditing PoPETs'26	Found 13 privacy bugs across 12 DP libraries by auditing their code.	COMPLEMENTARY	They audit the library's code; we audit the shipped release from the artifact alone — the layer they cannot see once the data leaves the building.
P8	McKenna — AIM VLDB'22	The state-of-the-art marginal DP synthesizer.	WE USE + EXTEND	We do NOT beat AIM at being AIM — we integrate it, reproduce its utility at full scale, and add the signed, checkable release layer it never had.
P7	Mohapatra — Missing Data VLDB'24	Studied whether missing data hurts utility or helps privacy.	WE EXTEND	We add amplification disclosure (RB10) and a bounded imputation audit on top of their setting.
P1	Stadler — Groundhog Day USENIX'22	Proved synthetic data leaks membership for outlier records via attacks.	WE PREVENT + CHECK	They demonstrate the attack; we make the channels it exploits (domain, seed, fingerprint) checkable and closeable at the artifact level.
P6	Ganev — SMOTE & Mirrors ICLR'26	Showed naive utility metrics miss real leakage.	WE GUARD	Our sanity-gate + membership check refuse to trust a bare 'no leak' — exactly their warning, operationalised.

4 · The one axis where we clearly lead

0 / 290

0 / 12

14

HuggingFace synthetic-data datasets that declare DP with an ϵ — none is signed or checkable

flagship real DP deployments (Apple, Google, US Census...) that are signed or machine-checkable

release-boundary checks (RB1–RB14) a third party runs from the artifact alone — the tool none of them ship

The honest headline: we are **not** claiming a better generator than McKenna's AIM or the Ganev group's audits — those are the state of the art and we build on them. We are the first to package a DP synthetic-data release as a **signed, machine-checkable label with a working checker** — exactly what Dibia et al. call for but never built, and what **every** real release fails today.

5 • What we deliberately do NOT claim

- Not a novel generator — AIM/MST are integrations of published methods; we reproduce their utility, we don't beat them.
- Not "audited $\epsilon = 0$ means no leakage" — that is the canary tool's detection floor; the honest reading is the operating range.
- Not "15/15 parameters" and not "we recover the original data" — neither is true; we detect what a release leaks beyond its ϵ .
- Full data helps correlation fidelity (Bank 32 $\times$, ACS 12 $\times$) but not uniformly at every ϵ — the DP domain grows with n .
- Multi-table auditing ships; multi-table synthesis is declared future work.

Generated by scripts/gen_explained_report.py from results/full/*, results/gdp_audit_*.json, research/wild_audit/. Per-paper verdicts grounded in research/18 (Claude-verified) and research/25 (prior-art). No number invented.